TECHNICAL DOCUMENTATION

nano-gpt

A Character-Level GPT (Transformer) From Scratch

A faithful PyTorch implementation of Andrej Karpathy's "Let's build GPT" lecture. Self-attention, multi-head, residual blocks and every part of a decoder-only language model.

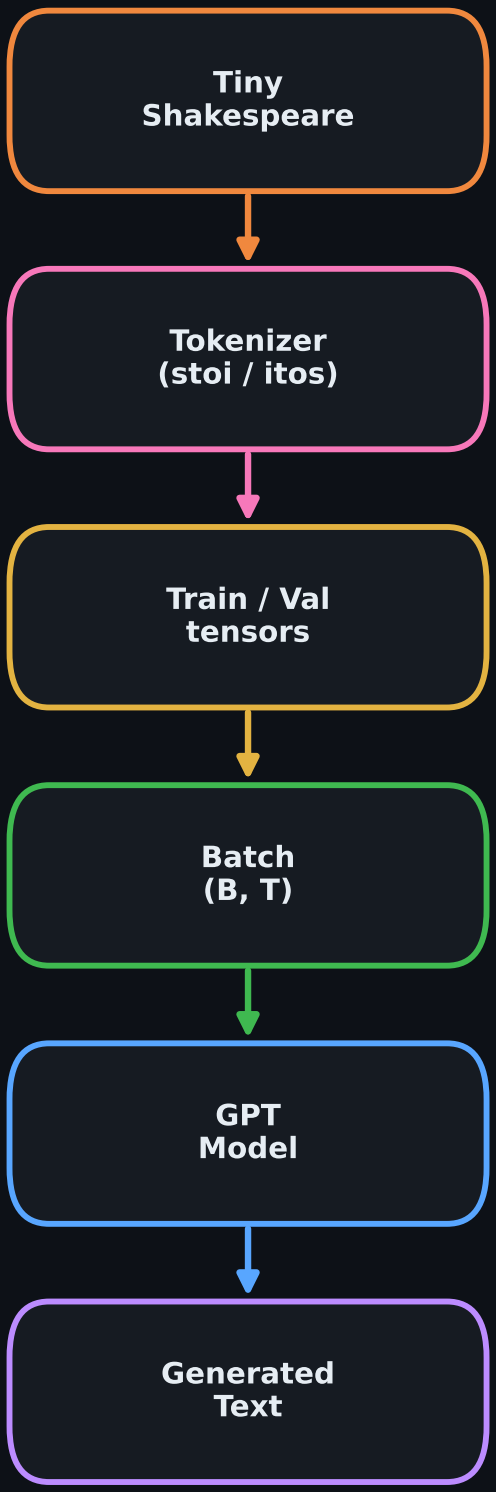
PyTorch · Transformer · Self-Attention · Char-level · Weights & Biases

Architecture at a Glance

Dataset	Tiny Shakespeare (~1.1 MB text)
Tokenizer	Character-level · vocab ≈ 65
Model	Decoder-only Transformer · 6 blocks · 8 heads
Embedding size	n_embed = 256 · block_size = 256
Training	AdamW + Cosine LR · Early Stopping · W&B

End-to-End Data Flow

The full pipeline, from raw text to freshly generated text. Each box maps to a function or module in the notebook.



`download_dataset()`

Downloads ~1.1 MB of raw text from Karpathy's char-rnn repo.

`build_tokenizer()`

Maps every unique character to an integer (encode / decode).

`make_splits()`

Converts text to a single long tensor, 90% train / 10% val.

`get_batch()`

Pulls (context, target) pairs from random starting points.

`BigramLanguageModel`

Embedding -> N x Transformer Block -> LayerNorm -> lm_head.

`model.generate()`

Autoregressive sampling generates new text character by character.

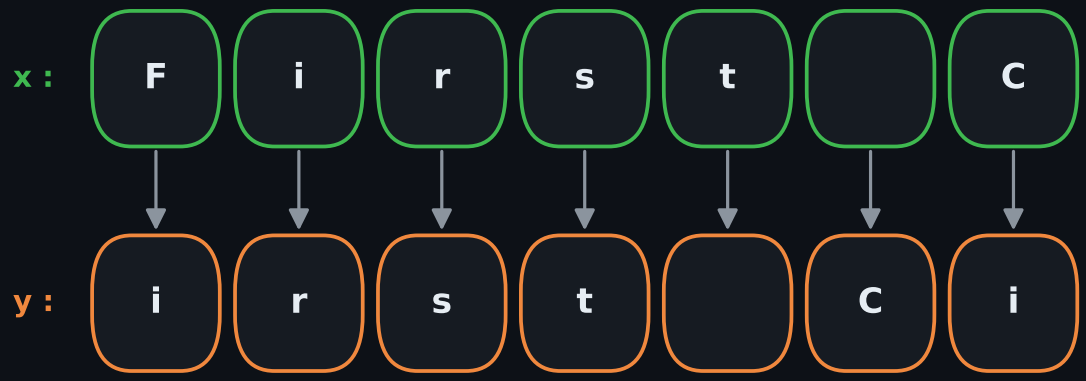
Tokenizer and Batch Construction

1 · Character-Level Tokenizer

Instead of BPE, each character is a single id. The vocab is tiny (~65), the embedding table is small, debugging is easy.

2 · get_batch → (context, target) pairs

A block_size-long x slice is paired with y, shifted one to the right. A single slice carries T independent training examples — this is the source of the Transformer's parallel learning.

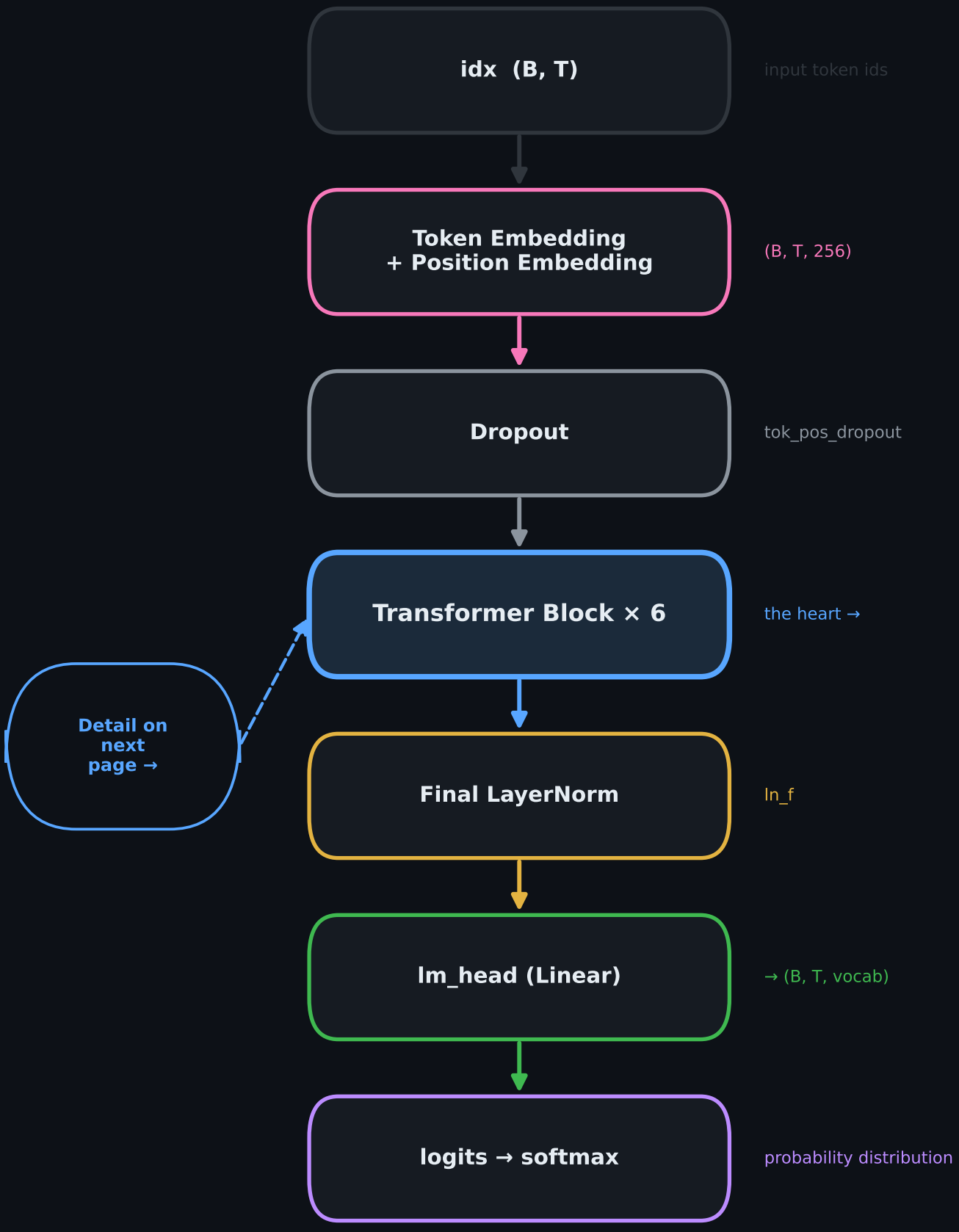


prediction direction: given x[:, t] the model predicts y[:, t]

Tensor Shapes

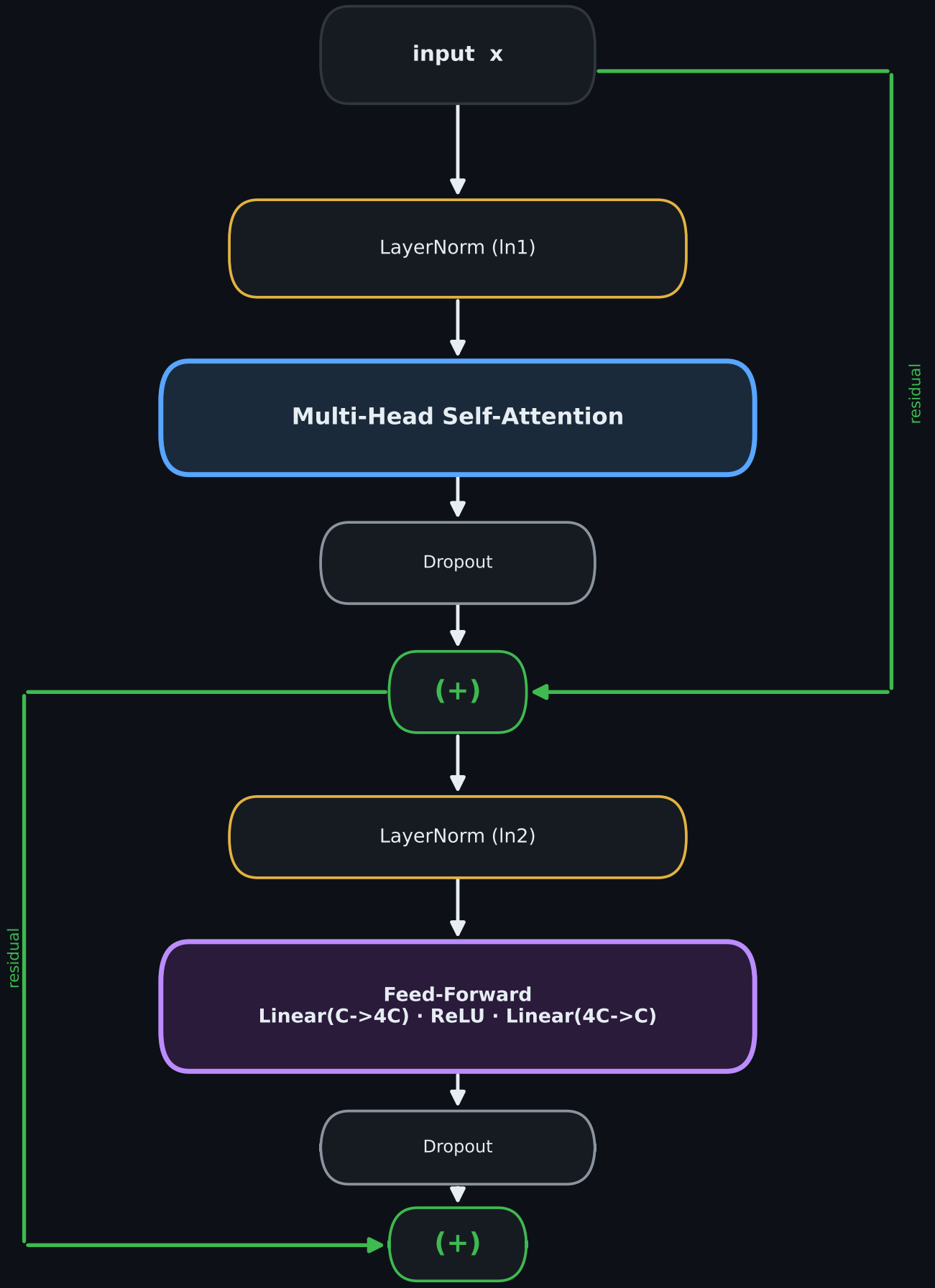
`x, y`	`(B, T)`	B = batch_size, T = block_size
`token_embedding`	`(B, T, C)`	C = n_embed
`logits`	`(B, T, vocab_size)`	score for each position
`loss`	`scalar`	cross-entropy

BigramLanguageModel — High-Level Flow



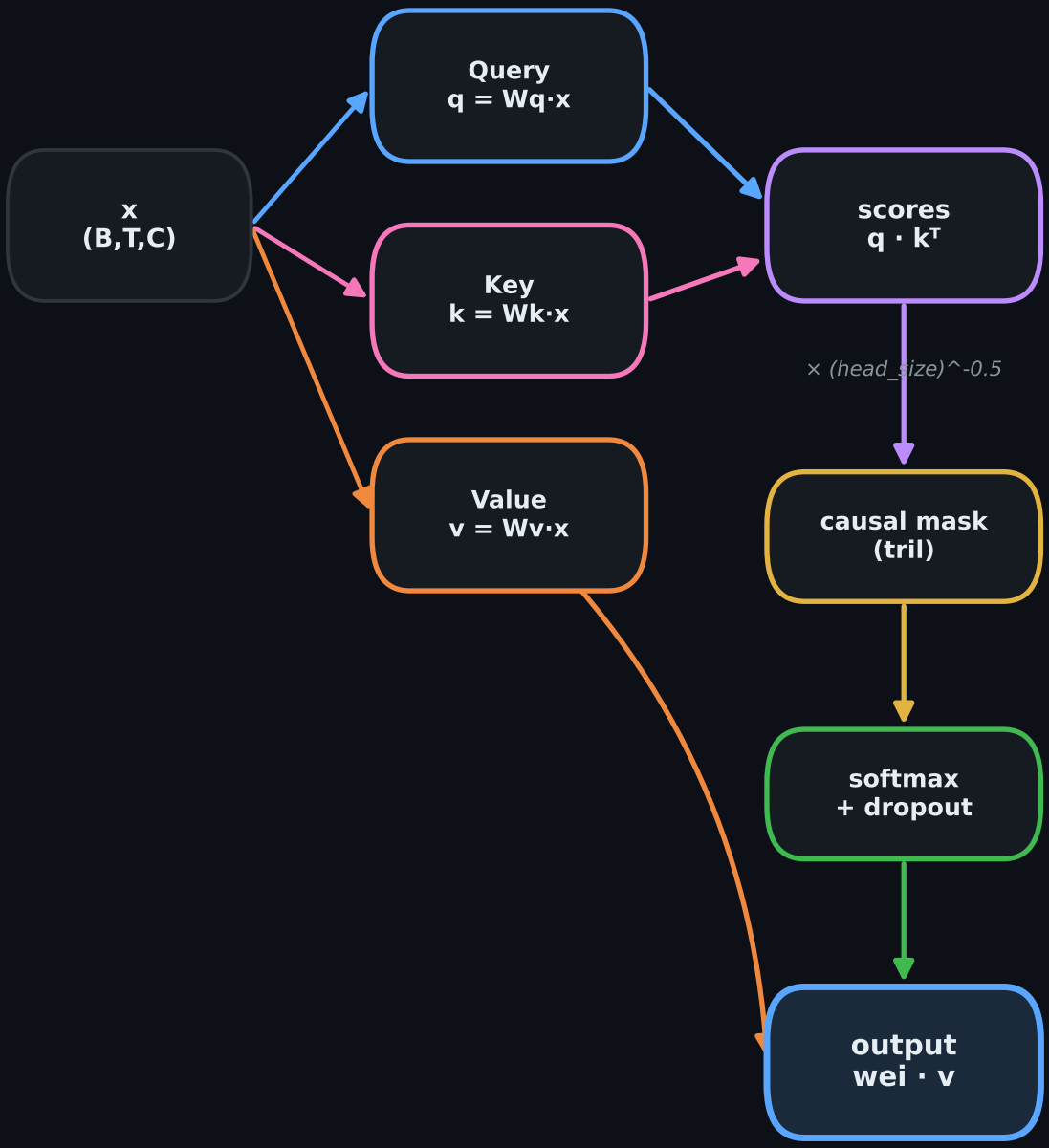
Transformer Block — Communicate + Compute

Each block has two sub-layers. Pre-LayerNorm is applied and the output of each sub-layer is added back to the input via a residual connection: x = x + Dropout(SubLayer(LN(x))).



A Single Self-Attention Head

Each head splits the input into three projections: Query, Key, Value. Scores are scaled, the future is hidden with a causal mask, softmax turns them into weights, Values are summed.



Causal mask: a token may only see itself and the past (future = -inf)

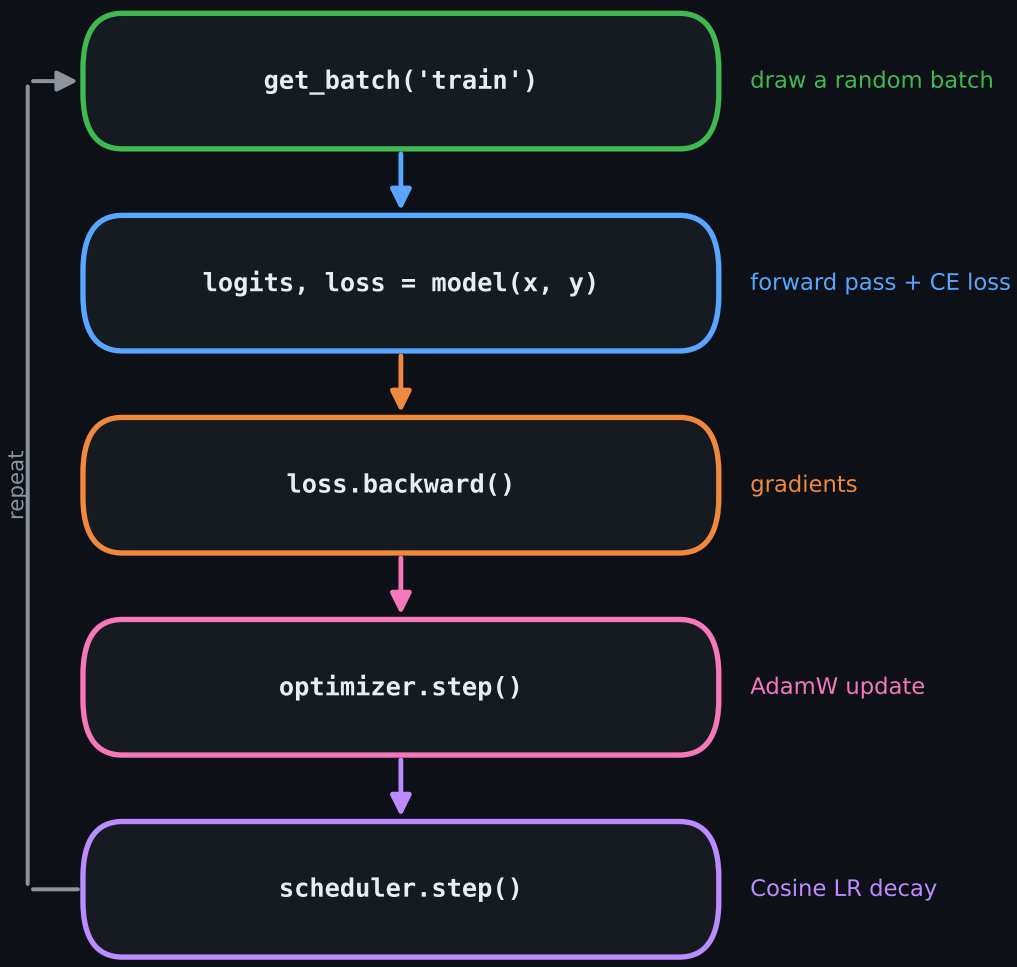
1	0	0	0	0
1	1	0	0	0
1	1	1	0	0
1	1	1	1	0
1	1	1	1	1

row = querying position
col = attended position

green (1) = allowed
dark (0) = masked

Training Loop and Hyperparameters

Iteration Loop



Regularization & Monitoring

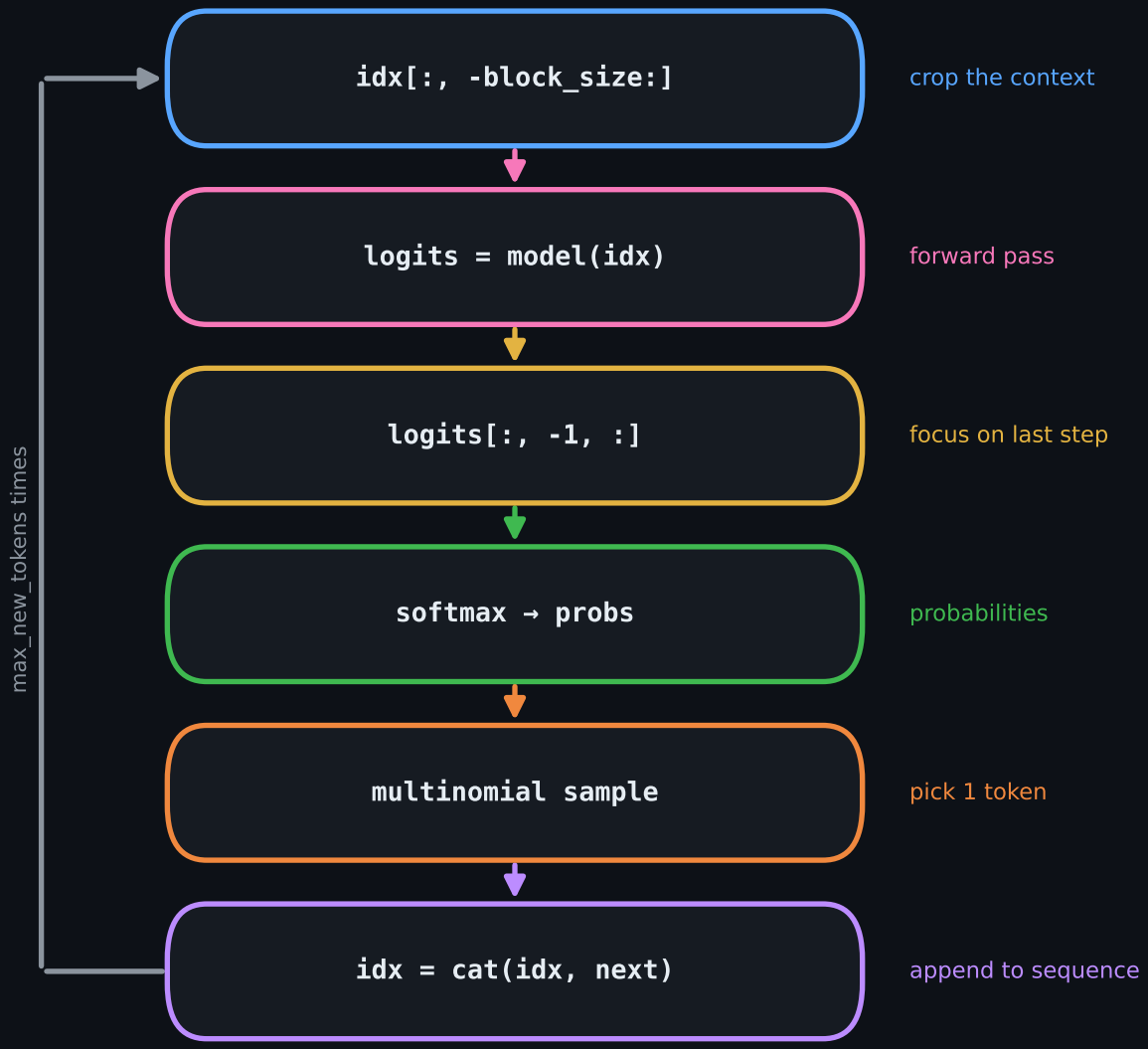
- **estimate_loss** — estimates train/val loss every 500 steps (eval mode)
- **Early Stopping** — stops if val loss stalls for 5 evaluations
- **Cosine LR** — lr 5e-4 -> 1e-5, smooth decay
- **Weights & Biases** — all metrics logged live

HERO RUN v4 — Configuration

block_size	256	batch_size	64	n_embed	256
n_head	8	n_layer	6	dropout	0.2
lr	5e-4	epochs	10000	weight_decay	1e-3

Autoregressive Text Generation

model.generate() takes the last block_size tokens as context at each step, samples one token from the last position's probability distribution and appends it. The loop feeds itself.



Why It Matters

This tiny model contains the exact same core mechanism as modern large language models (GPT): token + position embedding, causal self-attention, multi-head, residual blocks and autoregressive generation. Apart from scale, the fundamental architecture is identical — which makes it an ideal way to learn "GPT from scra